Age-Invariant Face Recognition Using Deep Learning

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***Abstract*—Age-invariant face recognition (AIFR) has garnered significant interest in computer vision due to the limitations of traditional face recognition systems in handling changes in facial appearance caused by aging. These changes can significantly affect performance in various applications, such as locating missing persons or verifying identities at airport security checkpoints, where available photos may be outdated. In this paper, our goal is to delve into AIFR by comparing the performances of three models trained on the same dataset using different methodologies: knowledge distillation [3], MTLFace [1], and ElasticFace [4]. Through knowledge distillation, we aim to compress a well-trained general face recognition (GFR) model to specialize in age-invariant face recognition, thereby improving computational efficiency while maintaining effectiveness in recognizing individuals across different ages. Our experimentation showed no significant improvement in performance when using the knowledge distillation model. We also found better performance in our knowledge distillation models when using the MS1M dataset over the B3FD dataset.**

Keywords*— AIFR (Age-invariant face recognition), knowledge distillation, MTLFace, ElasticFace, OursKD (our knowledge distillation network), MS1M, B3FD, Casia, AgeDB

I. Introduction

Age-invariant face recognition (AIFR) has gained significant interest in the field of computer vision, motivated by the need to accurately identify individuals across a wide age range. Traditional face recognition systems often struggle with changes in facial appearance changes in facial features [1]. AIFR aims to develop methods that can accurately recognize faces despite age-related changes, by either minimizing the influence of age progression on facial features identification or synthesizing faces to appear as if they are from the same age group [1]. Developing a high-performance AIFR model is not only technically challenging – given the complexity of aging processes and the difficulty in obtaining large, age-diverse training datasets – but is also of great importance for applying face recognition technology in real-world scenarios, where aging is a common and inevitable factor.

The successful implementation of AIFR technology can dramatically enhance security protocols and service delivery across multiple sectors. It significantly boosts the capabilities of surveillance systems, ensuring that individuals can be accurately identified over extended periods, which is essential in scenarios such as criminal investigations or monitoring public spaces. One key application is during immigration and border control. AIFR streamlines the immigration process by providing quick and reliable identity verification, thus enhancing both security and operational efficiency.

Even though AIFR can be very beneficial in different sectors, it is important to consider the potential of AIFR used in surveillance and tracking. This raises significant privacy concerns, as individuals may be continuously monitored without their consent. There are worries that this is already happening, and these improvements in AIFR could worsen the issue [2]. The technology could be employed for unauthorized mass surveillance, potentially leading to a loss of anonymity in public spaces. It is essential to establish clear ethical guidelines and regulatory frameworks to govern the use of AIFR, ensuring that it is used responsibly and with respect for individual privacy rights.

In this study, we aim conduct a comparative analysis of three distinct models: a knowledge-distilled model, a multi-task learning framework known as MTLFace, and the ElasticFace model. Each model is trained on the same dataset but utilizes different methodologies to achieve age-invariant recognition capabilities.

The knowledge distillation approach focuses on creating a compact and computationally efficient model by transferring the "knowledge" from a pre-trained General Face Recognition (GFR) model to a smaller model. This process is designed to retain the effectiveness of the original GFR model while specializing in recognizing individuals across various ages with reduced computational resources [3].

The MTLFace model, as proposed by Huang et al [1], is trained using a multi-task learning framework that simultaneously performs face recognition and age estimation. By integrating these tasks, MTLFace is inherently designed to adapt to the effects of aging on facial features, potentially improving its robustness in AIFR tasks.

Additionally, we plan to incorporate the ElasticFace model into our evaluation. ElasticFace introduces an elastic margin loss for deep face recognition, which allows flexibility in the push for class separability during training [4]. This model aims to enhance the discriminative power of face recognition models by utilizing random margin values drawn from a normal distribution in each training iteration, which could potentially improve performance in AIFR scenarios.

All three models—the knowledge-distilled model, MTLFace, and ElasticFace—will be assessed based on their accuracy in predicting unseen data from various datasets. The accuracies obtained will serve as a benchmark to determine the effectiveness of knowledge distillation as a model compression technique and to evaluate the performance of MTLFace and ElasticFace.

Also, training and testing models on different datasets is crucial for evaluating their robustness and generalizability. This approach creates an understanding of how well a model trained on one set of data can perform on another, unseen, set of data. By using MS1M-ArcFace and B3FD for training, followed by CASIA, AgeDB, and B3FD for testing we can understand which dataset is best for training AIFR models.

The major contributions of this paper are:

1. A detailed evaluation is done on MTLFace to determine its accuracy in AIFR.
2. We propose using knowledge distillation as an alternative to MTLFace or ElasticFace and comparing the performance of all methods.
3. We compare the performance of both model architectures when trained on the Microsoft Celeb (MS1M) and Biometrically Filtered Famous Figure Dataset (B3FD) datasets to determine which dataset is better for training in AIFR tasks.

The rest of the paper is broken down into 3 additional sections, the details of the sections are:

1. Section 2 – Discussion of datasets.
2. Section 3 – Breakdown of methodology, our knowledge distillation networks (OursKD) architecture, and experiment setup.
3. Section 4 – Summary of results and future direction of work.

II. DATASETS

This section covers the two datasets used throughout the experiments. The most popular dataset for facial recognition tasks is the MS1M dataset which contains 10 million face images. The second more modern dataset is the B3FD set, which contains 375,000. B3FD has significantly less images because unsupervised biometric filtering was used to remove incorrect entries and improve the quality and reliability of its age and gender. The following subsections provide the details of each dataset, the distribution of data, and how the data was used in our task.

A. M1SM Dataset

The MS-Celeb-1M (MS1M) dataset, created by Microsoft Research, is one of the largest publicly available face recognition datasets, containing approximately 10 million images of nearly 100,000 individuals [5]. This dataset was.

created by scraping information from the internet with the aim of developing face recognition technologies.

The vast number of images makes MS1M ideal for training deep learning models that require large amounts of data to achieve high accuracy. However, the dataset has been criticized for containing a lot of noise, such as mislabeled images and duplicates, which can adversely affect the training process [6].

The dataset is used for training the models in our experiment because of its extensive coverage and diversity, which includes a wide range of facial expressions, poses, and lighting conditions. This data set is not used as an unseen data set for testing in our experiment.

B. B3FD Dataset

The B3FD (Biometrically Filtered Famous Figure Dataset) is a facial image dataset specifically curated for age estimation tasks. It is derived from the IMDB-WIKI and CACD datasets and has been refined using unsupervised biometric filtering methods to remove incorrect samples [6]. The B3FD dataset contains 375,592 facial image samples of 53,759 unique subjects, averaging 6.99 samples per subject. The age labels range from 0 to 100 years.

The primary advantage of the B3FD dataset is its focus on high-quality, age-labeled images. The filtering process ensures that the dataset is free from noise present in the MS1M dataset, thereby improving the reliability of AIFR models trained on it [6]. A potential drawback is that the filtering process results in a reduction of the dataset size, with 53% of samples removed from the IMDB-WIKI data and 20% from the CACD data. This means that while the dataset is of higher quality, it offers fewer samples than the original datasets from which it was derived. This dataset is also used to train the models in our experiment as a comparison and is used as a test set as well in some of the trials.

C. CASIA-WebFace Dataset

The CASIA-WebFace dataset contains 494,414 images of 10,575 subjects [7]. It is known for its real-world applicability and diversity. One limitation is its smaller size like the B3FD dataset. CASIA-WebFace is often used for benchmarking and comparing face recognition algorithms due to its widespread recognition and acceptance in the research community. This dataset is not used to train any of the models but is used to test the performance of the models on unseen data.

D. AgeDB Dataset

The AgeDB dataset is specifically designed for testing and improving the age-invariance of face recognition systems [8]. It includes images specifically labeled with age, making it ideal for developing and testing age estimation algorithms. Again, it has a relatively small sample size, which can lead to lower accuracies in the final model. AgeDB is particularly useful age-related variations in facial recognition, providing a targeted dataset for this purpose. It contains 16,488 images with age labels ranging from 1 to 101 years. Similarly to Casia, this data set is not used to train any of the models but is used to test their performance on unseen data.

III. METHODOLOGY

This section outlines the architecture of the OursKD model, depicted in the figure. Initially, a facial image serves as input, which is then processed through both the teacher and student backbones to generate feature maps. Next, the student's feature map is inputted into the student model's head, producing the student's logits—a vector of probabilities that determine the class to which the input facial image belongs.

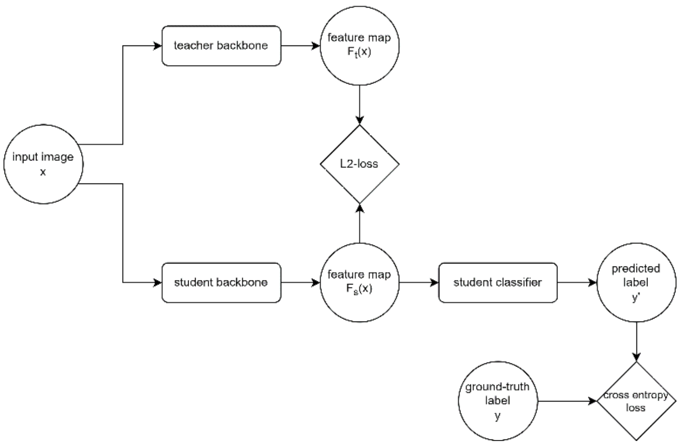

Figure 1: Architecture of the OursKD model.

A. Loss Functions

In the OursKD model, our goal is to transfer knowledge from the teacher model to the student model, aiming for a superior student model that leverages the strengths of both. To achieve this, we link the teacher with the student by the loss function, enabling the student to assimilate knowledge from the teacher during each update of the neural network's weights. Consequently, we have the following configurations:

For a facial image represented as x belonging to the set X, it is associated with a one-hot label y. When input this image x into the teacher's backbones, we obtain its feature maps denoted as $F_t(x)$, and similarly, feeding this image x into the student's backbones yields its feature maps denoted as $F_s(x)$.

If we view the knowledge transfer process as akin to teaching, then the input x represents the question the teacher aims to teach, and the feature map $F(x)$ represents the answer to that question. Therefore, we seek the student's answer to be as similar as possible to the teacher's answer. To achieve this, we minimize the difference between $F_t(x)$ and $F_s(x)$ by applying the $L2$ loss, which measures the Euclidean distance between the output feature maps of the teacher and student's backbones.

Besides, as the teacher's feature map $F_t(x)$ may not perfectly represent the facial image x, we take precautions against the student's model overfitting to the teacher's model. To achieve this, we pass the output feature map of the student's backbone through the student's classification head, obtaining the probability vector $H_s(F_s(x))$ and then apply cross-entropy between the ground-truth label y of image x and this probability vector.

B. Preformance metrics

When evaluating the OursKD model, it's insufficient to only consider the prediction accuracy of the student's model, which is easily calculated by determining the percentage of correct predictions. We also aim to evaluate how effectively the output feature map of the student's backbone represents the input image. Thus, we need to integrate these two aspects, and for that purpose, we have devised a unique method within our model.

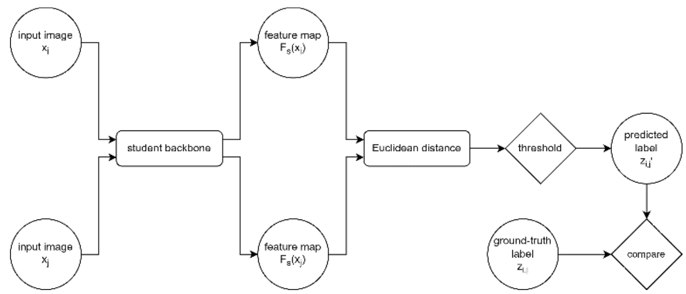

Figure 2:

We employed two distinct facial images, x_i and x_j, as inputs along with a label $z_{i,j}$ that signifies whether they pertain to the same individual. When $z_{i,j}$=0, it denotes different facial classes; conversely, $z_{i,j}$=1 signifies the same facial class with varying ages.

Next, we compute the Euclidean distance between the two feature maps $F_s(x_i)$ and $F_s(x_j)$, and then select a threshold that distinguishes whether two feature maps correspond to the same identity. Setting the threshold to zero isn't viable because facial images of the same identity but different ages can generate similar but not identical feature maps. Hence, we determine the optimal threshold that maximizes accuracy.

We hypothesize that if the Euclidean distance between the two feature maps aligns with the best threshold we've selected, then these two images belong to the same facial class but with different ages. Otherwise, they pertain to entirely different individuals.

IV. Results and Discussions

A. Results

TABLE I. Individual Models Performance

Model	**Train Set**	**Verification Accuracy**		
		B3FD	***AgeDB***	***CASIA***
ElasticFace	B3FD	91%	57%	68%
MTLFace	B3FD	96.3%	77.4%	82.2%
ElasticFace	MS1MV2	87.1%	98.3%	96.1%

TABLE II. Knowledge-Distillation Models Performance

Model		**Verification Accuracy**		
Student	***Teacher-MS1M***	***B3FD***	***AgeDB***	***CASIA***
ElasticFace-MS1M	ElasticFace	95.6%	74.8%	79.9%
ElasticFace-B3FD	ElasticFace	95.7%	76.5%	82.1%

MTLFace-scratch	ElasticFace	83.4%	59.6%	67.3%
MTLFace-B3FD	ElasticFace	96.0%	76.5%	82.6%

TABLE III. ADDITIONAL KD MODELS PERFORMANCE

Model		Verification Accuracy		
Student	***Teacher-BSFD***	***B3FD***	***AgeDB***	***CASIA***
ElasticFace-B3FD	ElasticFace	90.8%	64.7%	71.8%
ElasticFace-B3FD	MTLFace	94.3%	79.7%	82.9%

To establish a baseline for our knowledge distillation models, we first evaluated the performance of ElasticFace and MTLFace on age-invariant face verification as individual models. We trained each model on the B3FD dataset from scratch, and then tested on a subset of B3FD (similar but unseen in training) and on the AgeDB and CASIA datasets (different and unseen in training). We also evaluated the original ElasticFace model – pretrained on MS1MV2 by Boutros et al. – on the unseen B3FD dataset.

As shown in Table I, ElasticFace trained on MS1M and MTLFace trained on B3FD show better generalization capability on unseen data. From this table, we concluded that the ElasticFace model pretrained on MS1M acquired the best knowledge (feature maps) for face verification and is thus the most suitable candidate for the teacher model in our knowledge distillation scheme.

Subsequently, we experimented with using either ElasticFace or MTLFace as the student model, training them alongside the teacher model on the B3FD dataset, both from scratch and from pretraining on the same dataset. By loading the student models with pretrained weights, we aimed to fine-tune the features obtained through training on B3FD with features the teacher obtained through training on MS1M. Each KD model was again tested on the B3FD testing subset, AgeDB, and CASIA for face verification. Table II captures verification accuracies from some of our noteworthy models. The best results come from using either ElasticFace or MTLFace pretrained on B3FD as the student and ElasticFace pretrained on MS1M as the teacher.

Additionally, as captured in Table III, we also experimented with using ElasticFace or MTLFace pretrained on B3FD as teachers. The former is outperformed by its counterpart pretrained on MS1M, while the latter shows no significant improvement over the individual MTLFace model, defying the purpose of using knowledge distillation.

B. Discussions

TABLE IV. COMPARISON OF INDIVIDUAL AND KD MODELS

Model		Verification Accuracy		
Student-B3FD	***Teacher***	***B3FD***	***AgeDB***	***CASIA***
ElasticFace	n/a	91%	57%	68%
ElasticFace	ElasticFace-B3FD	90.8%	64.7%	71.8%
ElasticFace	ElasticFace-MS1M	95.7%	76.5%	82.1%
ElasticFace	MTLFace-B3FD	94.3%	79.7%	82.9%

For a fair comparison, we initially trained an ElasticFace model from scratch on the B3FD dataset alone, and subsequently observed performance improvements when the model was trained alongside a teacher. The choice of the teacher model appears to significantly affect the student's performance, as summarized in Table IV. On one hand, improvements are more notable when there is a disparity between the student and teacher models, either in their structures or in the datasets they were pretrained on, indicating that the student can combine different extracted features and become more robust to variations and biases. On the other hand, using a model pretrained on MS1M as teacher led to considerably better results compared to the one pretrained on B3FD. This discrepancy could be attributed to the significant difference in the scope of each dataset. Since MS1M contains many more samples than B3FD, it allows the teacher model to learn a more diverse set of features, which in turn helps the student model generalize well on unseen data.

Additionally, as shown in Table II, loading pretrained weights into the student model and the specific choice of those weights also impacts training results. With the same teacher model (ElasticFace pretrained on MS1M), the student MTLFace model achieved much higher verification accuracies when loaded with pretrained weights. Similarly, the student ElasticFace model showed slight improvements when it used pretrained weights that differed from the teachers. These results further support our hypothesis that knowledge distillation can enable models to fine-tune or combine features learned from different architectures or datasets.

However, there are certain limitations to our approach. First, it should be noted that we did not optimize the hyperparameters and other settings specifically for training ElasticFace on B3FD, such as learning rate or batch size, which could have led to better baseline and thus altered comparison results.

Furthermore, even though our KD models showed improvements compared to the individual ElasticFace trained on B3FD, the increase in verification accuracies was insignificant or even negative compared to the individual MTLFace trained on B3FD, especially for models that used MTLFace as the student. We consider this to be an indicator of the overall failure of our KD models. One possible cause could be flaws in our training scripts, as we did not fine-tune the weights of each loss term (L2 loss between teacher and student feature maps, cross-entropy loss, and triplet loss between student output and true label) for the MTLFace model, leading to potentially suboptimal or even conflicting effects.

V. CONCLUSIONS

Our experimentation in knowledge distillation techniques in the context of age-invariant face recognition (AIFR) has demonstrated mixed results. While the use of a teacher model pretrained on a more diverse dataset (MS1MV2) has shown significant enhancement in the performance of student models, particularly when preloaded with pretrained weights. However, the overall effectiveness of knowledge distillation remains inconsistent. The experiments reveal that the choice of teacher model and the disparity in dataset scope play crucial roles in the success of the student models in generalizing to unseen data. Despite some models showing promising improvements, the approach did not consistently outperform the baseline models, particularly when MTLFace was used as a student. This suggests that while knowledge distillation holds potential for improving AIFR systems, further refinement in model training strategies, optimization of loss functions, and a deeper understanding of feature transfer are necessary to fully leverage its capabilities.

Overall, we found that training the models using the knowledge distillation network had better results when trained with the MS1M dataset. This was most likely the case because having more data points allowed for more training even though some of the data was flawed.

In future work, there could be more research on ways to transfer learned features more effectively for models to generalize better on unseen data. Exploring the substantial performance discrepancies between ElasticFace trained on B3FD verses on MS1M could also provide valuable insights. Further investigations could also focus on refining the choice of loss terms for each specific model and dataset to enhance their effectiveness. Finally, expanding the range of datasets and incorporating real-world scenarios could provide a more comprehensive assessment of the models' practical applicability and robustness in diverse environments.

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